MIL-T-12220C(ME)
2 October 1973
SUPERSEDING
MIL-T-12220B(ME)
7 July 1967

MILITARY SPECIFICATION

TRESTLE, RAILWAY: STEEL UNIT, LIGHT STANDARD (L-TYPE)

1. SCOPE

1.1 This specification covers the structural components of the light standard (L-type) unit, steel, railway trestle.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

FF-B-575	- Bolts, Hexagon and Square.
FF-N-836	- Nut: Square, Hexagon, Cap, Slotted, Castle, Knurled, Welding and Single Ball Seat.
FF-W-92	- Washers, Metal, Flat (Plain).
QQ-S-781	- Strapping, Steel, Flat and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-C-650	- Crates, Wood, Open and Covered.

Military

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation-Packaging, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-3774	- Crates, Wood; Open, 12,000-and 16,000-Pound Capacity.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.

FSC 5420

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MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.

DRAWINGS

ME

D2623-1 through -4	- Light Standard Unit Steel Trestle.
C13206E4482	- Washer, Beveled.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)

A5.1	- Mild Steel Covered Arc-Welding Electrodes.
B3.0	- Standard Qualification Procedure.
D1.1	- Structural Welding Code.

(Application for copies should be addressed to the American Welding Society, 2501 Northwest Seventh Street, Miami, FL 33125.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A36	- Structural Steel.
A108	- Cold-Finished Carbon Steel Bars and Shafting.
A120	- Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless Steel Pipe for Ordinary Uses.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street N. W., Washington, DC 20036.)

3. REQUIREMENTS

3.1 Description. The trestle components shall be fabricated as shown on Drawings D2623-1 through -4 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish one of each trestle component as applicable, for inspection within the time frame specified (see 6.2) to prove prior to starting production, that his production methods will produce trestle components that comply with the requirements of this specification. Examination and test shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.3.1 Structural steel. Structural steel shall conform to ASTM A36.

3.3.2 Steel bars and rods. Steel bars and rods shall conform to ASTM A108, grade 1018 to 1020.

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3.4 Welding electrodes. Welding electrodes shall conform to AWS A5.1, classification E60 or E70.

3.5 Pipe. Pipe shall conform to ASTM A120, standard weight (schedule 40), except that ends shall not be threaded and the hydrostatic test will not be required.

3.6 Bolts. Bolts shall conform to FF-B-575, type II, steel.

3.7 Nuts. Nuts shall conform to FF-N-836, type II, style 4, carbon steel.

3.8 Washers. Standard washers shall conform to FF-W-92, type A, grade I, class A. Beveled washers shall conform to Drawing C13206E4482, dash No. 14.

3.9 Identification marking. The trestle components, except bolts, nuts, and washers, shall be identified as shown on the drawings. The bolts, nuts, and washers shall be identified in accordance with MIL-STD-130.

3.9.1 Metal stamping. The trestle components shall be metal stamped as shown on the drawings. Metal stamping shall be of such a depth as to be legible after painting.

3.9.2 Stenciling. The nomenclature, Federal stock number, weight, and cubage shall be stenciled near the center on one side of each trestle component except bolts, nuts, and washers. The letters "NET WT" shall precede the numerals in the weight and the letters "CU" shall precede the cubic displacement. Markings shall be Gothic type capitals and Arabic numerals 2 inches high where space permits. On small components, the stenciling shall be as large as practicable. Stenciling shall conform to MIL-T-704.

3.10 Treatment and painting. All trestle components except bolts, nuts, and washers shall be treated and painted in accordance with MIL-T-704, type A.

3.11 Workmanship.

3.11.1 Steel fabrication. Steel used in the fabrication of trestle components shall be free from laminations and bends. Bent material shall be straightened by methods that will not change the physical properties of the material. Steel may be sheared, sawed, or flame cut. Burned surfaces of flame-cut material shall be ground or machined to remove ash and cooling checks. All burrs and projections shall be removed.

3.11.2 Bolted connections. Boltholes may be subpunched or drilled undersize prior to welding and reamed to full size after all welding has been completed, or shall be drilled to full size after all welding has been completed. Boltholes shall be drilled or reamed straight through from one side of the component.

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3.11.3 Welders and welding.

3.11.3.1 Welders. Before assigning any welder or automatic welder equipment operator to welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder or operator has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society (B3.0).
Welding Qualification of the ASME Boiler and Pressure Vessel Code (Section IX).

3.11.3.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the provisions of the American Welding Society publication "D1.1-Structural Welding Code". All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed, and the deposit and adjoining base metal shall be wire brushed clean.

3.11.4 Jigs and fixtures. Shop-fabricated components shall be assembled in jigs or fixtures and joined while held in position. Jigs or fixtures shall be designed to prevent distortion or to limit the distortion to an amount that can be removed by straightening and stress relieving without changing the physical characteristics of the metal.

3.11.5 Tolerances. The location of all holes shall be within plus or minus 1/32 inch of the dimensions shown on the drawings, and the diameters of all holes shall be within plus 0.010 inch and minus 0.005 inch of the specified diameter. The tolerances regarding the positioning of holes, the thickness of material, and the length, width, and depth of members shall be referenced from one side or end of the component and shall not be cumulative.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification

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where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.5).

4.3 Preproduction inspection.

4.3.1 Examination. The trestle components shall be examined as specified in 4.4.2. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the following defects. AQL shall be 2.5 percent defective.

101. Materials not as specified.
102. Dimensions not as shown on drawings.
103. Components missing or not as specified.
104. Weld defects, such as base-metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, porosity, lack of penetration, slag inclusions, or spatter.
105. Components containing sharp edges or projections.
106. Treatment and painting not as specified.
107. Identification or special markings incorrect, illegible, or missing.

4.5 Inspection of preparation for delivery.

4.5.1 Preproduction pack inspection.

4.5.1.1 Examination. The preproduction pack shall be examined for the defects specified in 4.5.2.3. Presence of one or more defects shall be cause for rejection.

4.5.1.2 Tests. The preproduction pack of the crated components for level A or B shall be subjected to the railroad car test as specified for guided-impact tests in accordance with MIL-STD-1186, appendix A. The car shall strike a string of five empty cars with draft gear extended and the brakes set, at a speed of not less than 10 miles or more than 11 miles per hour. At the conclusion of the preproduction pack test, the pack shall be examined.

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Any shifting of contents, loosening or breaking of ties, stays, blocking, or bracing, or any visual damage to the components shall constitute failure of the preproduction pack and shall be cause for rejection.

4.5.2 Quality conformance inspection of pack.

4.5.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.2.3 Examination. Samples selected in accordance with 4.5.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

108.	Materials, methods, and containers not as specified for level A or B. Each incorrect material, method, or container shall constitute one defect.
109.	Unpainted metal surfaces not coated with preservative for level A or B.
110.	Bundling and securing not as specified for level A or B.
111.	Strapping not spaced as specified for level A or B.
112.	Boxed contents not secured within the box in a manner to prevent free movement for level A or B.
113.	Blocking, bracing, and anchoring of crated components not in accordance with the appendix to the crate specification and the referenced document for level A or B.
114.	Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preproduction pack. The supplier shall furnish a preproduction pack for examination and test within the time frame specified (see 6.2) to prove, prior to starting production packaging that the applied preservation, packaging, packing, and marking comply with the preparation for delivery requirements of this specification. Examination and test shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The preproduction pack may be accomplished utilizing either the preproduction trestle or trestle components or a production trestle or production trestle components. If the preproduction trestle is utilized, any preservation, packaging, and packing shall be removed by the supplier at no expense to the Government when requested by the Government to facilitate comparison between the preproduction trestle and the production trestle.

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5.2 Preservation and packaging. Preservation and packaging shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Unprotected surfaces. Unpainted surfaces of trestle components shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.2.1.2 Column units (Mk L-1, L-2, L-3, L-4). Column units of like description shall be bundled by bolting the flanges together with 3/4-inch-diameter bolts and shall be strapped with strapping conforming to QQ-S-781, type I, finish B, 3/4 inch by 0.035 inch, as shown in figure 1.

5.2.1.3 Struts (Mk L-5). Struts (Mk L-5) shall be bundled in quantities most convenient for packing but not to exceed 12 each. The struts shall be nested and strapped with not less than two straps conforming to QQ-S-781, type I, finish B, 3/4 inch by 0.035 inch. The straps shall be placed not more than 30 inches apart.

5.2.1.4 Bracing angles (Mk L-6, L-7). Bracing angles of like description shall be bundled in quantities most convenient for packing but not to exceed 20 each. The angles shall be nested and strapped with not less than two straps conforming to QQ-S-781, type I, finish B, size as applicable. The straps shall be placed not more than 30 inches apart.

5.2.1.5 Trestle joist units (Mk L-8, L-9). Trestle joist units marked L-8 shall be bundled as shown in figure 2. Trestle joists marked L-9 shall be bundled in a similar manner as shown in figure 2.

5.2.1.6 Splice plates (Mk L-10, L-11), shim plates (Mk L-12), and filler plates (Mk L-35, L-36, L-37, L-38, L-58, L-59). These items of like description shall be stacked and bundled in quantities most convenient for packing. Each bundle shall be secured with not less than two 3/4-inch-diameter bolts through matching holes.

5.2.1.7 Derrick masts (Mk L-62). Derrick masts shall be bundled in quantities most convenient for packing but not to exceed six each. The masts shall be nested and strapped with not less than two straps conforming to QQ-S-781, type I, finish B, 3/4 inch by 0.035 inch, not more than 30 inches apart.

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5.2.1.8 Bracing plates (Mk L-13), grillage joist clips (Mk L-14), universal joist clips (Mk L-56), adapters (Mk L-77), bracing members (Mk L-57), rakers (Mk L-17), spacers (Mk L-30, L-15, L-16, L-55B, L-55C), derrick brackets (Mk L-60), holding plates (Mk L-61), fork and sleeve (Mk L-66), derrick jib pins (Mk L-40), and bolts, nuts, and washers. These items of like description shall be packaged in close-fitting boxes conforming to PPP-B-621, class 2, style 2, or in boxes conforming to PPP-B-601, overseas type, style A or B. Blocking, bracing, or anchoring required to prevent free movement of the contents within the box shall be provided. The boxes shall be closed in accordance with the appendix to the applicable box specification. Strapping shall be zinc coated.

5.2.2 Level B. Preservation and packaging shall be as specified in 5.2.1 for level A except boxes shall be domestic type or class as applicable and strapping is not required to be zinc coated.

5.2.3 Level C. The components of the trestle shall be preserved and packaged to afford protection against deterioration and damage from the supplier to the initial destination.

5.3 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.3.1 Level A. All packaged and bundled components of like description, except those components specified in 5.2.1.2, 5.2.1.5, and 5.2.1.8, shall be packed together in boxes conforming to PPP-B-621, class 2, style 2, or PPP-B-601, overseas type, style A or B, or in crates conforming to PPP-C-650, type V, style A, class 1, MIL-C-3774, nailed assembly, or MIL-C-104, type I, class 1 or 2, style A; as applicable to the weight of the components to be packed. Blocking, bracing, and anchoring of the components within the box or crate shall be in accordance with MIL-STD-1186. Closure and strapping shall be in accordance with the applicable specification or appendix thereto. Strapping shall be zinc coated. Packaged and bundled components specified in 5.2.1.2, 5.2.1.5, and 5.2.1.8 shall not require overpacking.

5.3.2 Level B. Packing shall be as specified in 5.3.1 for level A, except boxes shall be domestic type or class as applicable and strapping is not required to be zinc coated.

5.3.3 Level C. The components of the trestle shall be packed to insure carrier acceptance and safe delivery to destination at lowest rates in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.4 Marking. In addition to any special marking required, bundles, packages, and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The trestle is intended for use as the structural support component for the light railway bridge.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction trestle components (see 3.2).
(c) Time frame required for submission of the preproduction pack (see 5.1).
(d) Level of preservation and packaging and level of packing required (see 5.2 and 5.3). Level B preservation and packaging is intended to provide limited but economical protection and should be specified only when it is determined the trestle will be held in covered storage.

6.3 Preproduction model. Any changes or deviations of production trestle components from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish trestle components conforming to this specification.

6.4 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to preserve, package, pack, and mark the trestles or trestle components, as applicable, in accordance with this specification.

Custodian:

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Preparing activity:

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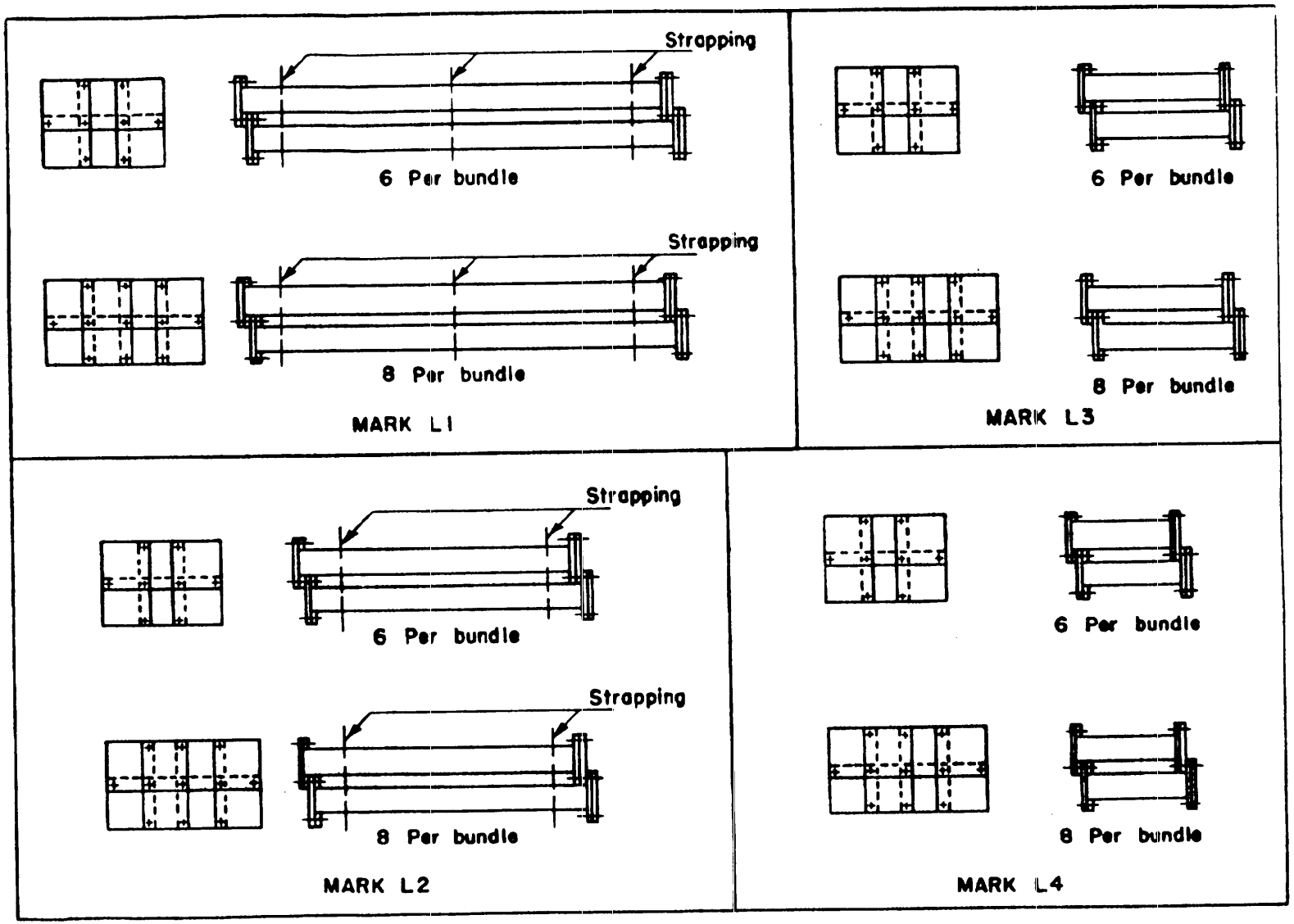


Fig. 1 Column units – Bundle arrangement

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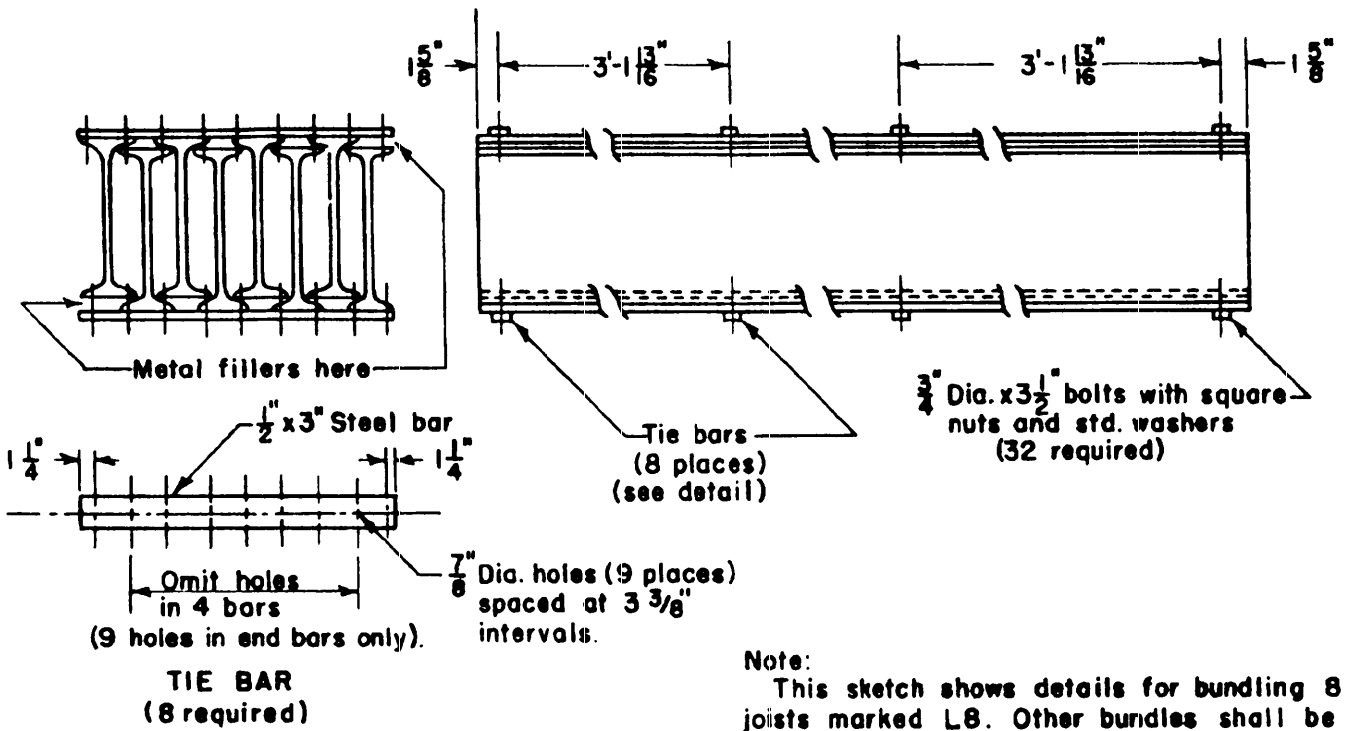


Fig. 2 Joists – Details of bundling.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: MIL-T-12220C(ME) Trestle, Railway: Steel Unit, Light Standard (L-Type)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426
11 Oct 72 (Rev)

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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